12-30 第2篇

阅读理解（说明文） Why Is a Yawn So Easily "Caught" from Other People?

① You are sitting quietly in a Grade-8 English class at 14:25 on a Monday afternoon. A classmate three desks in front of you **yawns softly**. Within about six seconds, **without knowing why**, your own mouth begins to open, your eyes water slightly, and a yawn of your own escapes — almost as if your classmate's yawn had **jumped across the classroom** into your body. Five of your classmates catch the same yawn in the next twenty seconds. Mr Qin, trying to talk about present perfect tense, shakes his head with a small smile. Most middle-school students have lived through this scene hundreds of times without ever asking a simple question: **why is yawning so easily passed from one person to another?** Scientists have, interestingly, been studying this since the 1950s, and although no single answer is yet 100% proven, four small facts are now widely agreed. This article explains them in simple words.

② **Fact 1 — A yawn is not really caused by too little oxygen.** For many years, the most popular idea among ordinary adults was that yawning "wakes the body up" by pulling in a big gulp of oxygen. Careful scientific experiments, however, have weakened this old idea. In a 1987 study at the University of Maryland, volunteers were asked to **breathe pure oxygen** for several minutes; their yawning rate did **not** go down. Volunteers were also asked to breathe air **rich in carbon dioxide**; their yawning rate did **not** go up. Clearly, **yawning is not mainly about oxygen levels in the blood**. What, then, is a yawn for? The current best answer is: **a yawn is a small brain-cooling action**.

③ **Fact 2 — A yawn may quietly cool the brain.** In the past ten years, Professor Andrew Gallup at Utica University has shown, in a careful set of studies, that **yawning gently drops the temperature of the brain by about 0.3 — 0.4 °C**. When you yawn, three small things happen, all at once. First, your jaw opens wide, which **stretches the blood vessels** in your cheeks and draws cool blood deeper into your head. Second, you take one deep breath of cooler outside air across the **wet inner walls** of your mouth and nose, creating a tiny cooling effect, like wind on a damp towel. Third, your throat muscles briefly press the **warm blood leaving your head** back toward the heart, making space for cooler blood coming up. Together, these three small actions lower the brain's temperature by a small but real amount — enough to **wake you up a little**, but not so much as to wake you up fully. This is why we yawn most often **before sleep, after sleep, and during boring long moments** — all times when the brain's temperature is slightly higher than is useful for clear thinking.

④ **Fact 3 — Catching a yawn is not a weakness; it is a small sign of empathy.** Here is the most interesting part. Studies on thousands of volunteers in eight countries have

shown that **about 60 — 70% of healthy adults catch a yawn within a minute of seeing another person yawn**; young children under four, and certain people with very low emotional empathy scores, catch the yawn much less often. Why? Because catching a yawn, it seems, is not a breathing problem — it is an **automatic social copying action**, done by a small brain area called the **mirror system** (镜像神经系统), the same area that lets you feel a small stab of sadness when you see somebody else's knee hit a door. In other words, a middle-school student who catches a yawn easily is, statistically, more likely to be a student who **gently notices other people's feelings**. So the next time you catch five yawns in one afternoon, do not be embarrassed — you are simply a person whose mirror system is healthy and quietly awake.

⑤ **Fact 4 — You can catch a yawn from a picture, a sound, and even the word "yawn".** Careful experiments have shown that you do not need to **see** a yawning person to catch a yawn. A **silent photograph** of somebody yawning causes about 40% of viewers to yawn within one minute. A **sound recording** of a yawn — with no video at all — causes about 30% to yawn. And in one small 2003 study in Britain, merely **reading the word "yawn" several times** on a page caused 22% of readers to yawn, even alone in a silent room. (You may, quite possibly, have already yawned while reading this paragraph.) The yawn reflex, it seems, is not tied to sight alone; it is tied to the **idea** of yawning itself, wherever that idea enters the brain.

⑥ **Fact 5 — Dogs catch yawns too, especially from their owners.** If more proof were needed that yawning is an empathy-linked behaviour, pet owners have kindly given it. A 2013 study at the University of Tokyo showed that **about 45% of domestic dogs yawn within a minute of watching their owner yawn**, and the rate rises to roughly **60% when the owner is the dog's main carer**. Dogs do not catch yawns from strangers as easily. This is one small reason why many scientists now believe that catching a yawn is, at its heart, a **cross-species social signal** — a quiet way living animals say, *"I am paying attention to you, and my body is, in a small way, syncing with yours."*

⑦ So the next time a classmate three desks ahead yawns softly on a Monday afternoon and you, six seconds later, feel your own mouth open without permission, you may quietly remember four small facts: (1) your classmate's yawn is very probably a gentle brain-cooling action, not an oxygen rescue; (2) your own catch-yawn is an automatic sign of a healthy, awake mirror system; (3) the yawn probably travelled to you through your eyes, your ears, or even the idea of yawning itself; and (4) your small shared yawn with five classmates is, in quiet biological truth, one of the oldest social signals on Earth, shared even by dogs. A yawn, in short, is far more interesting than most adults believe. **The next time you catch one, give it a small thank-you — it is proof that your body is, quietly, still listening to the bodies around you.**

注释： yawn 打哈欠；oxygen 氧气；carbon dioxide 二氧化碳；jaw 下巴；empathy 共情；mirror system 镜像神经系统；domestic dog 家犬

根据短文内容，选择最佳答案。

1. According to Fact 1, what evidence weakens the old idea that yawning is mainly about oxygen?

A. The old idea has never been tested in any scientific study.
B. New studies have fully confirmed that yawning is entirely about oxygen and carbon dioxide.
C. For many years the most popular idea among ordinary adults was that yawning "wakes the body up" by pulling in a big gulp of oxygen, but careful scientific experiments have weakened this old idea: in a 1987 study at the University of Maryland volunteers asked to breathe pure oxygen for several minutes showed no drop in yawning rate, and volunteers asked to breathe air rich in carbon dioxide showed no rise in yawning rate; clearly yawning is not mainly about oxygen levels in the blood, and the current best answer is that a yawn is a small brain-cooling action.
D. The 1987 study proved that yawning doubles oxygen intake.

2. According to Fact 2, what three small actions inside the body cool the brain during a yawn?

A. A yawn increases the heart rate enormously for one full minute.
B. In the past ten years Professor Andrew Gallup at Utica University has shown in careful studies that yawning gently drops the temperature of the brain by about 0.3—0.4°C by three small actions all at once: first, the jaw opens wide and stretches the blood vessels in the cheeks, drawing cool blood deeper into the head; second, one deep breath of cooler outside air crosses the wet inner walls of the mouth and nose, creating a tiny cooling effect like wind on a damp towel; third, the throat muscles briefly press the warm blood leaving the head back toward the heart, making space for cooler blood coming up; together these lower brain temperature by a small but real amount — enough to wake a person a little but not fully — which is why we yawn most often before sleep, after sleep, and during boring long moments, all times when brain temperature is slightly higher than useful for clear thinking.
C. The brain is cooled only by drinking cold water during a yawn.
D. Gallup's research showed that yawning warms the brain by 3°C.

3. What is the writer's interpretation of Fact 3 about catching yawns?

A. Studies on thousands of volunteers in eight countries have shown about 60—70% of healthy adults catch a yawn within a minute of seeing another person yawn, while young children under four and certain people with very low emotional empathy scores catch the yawn much less often; catching a yawn is not a breathing problem but an automatic social copying action done by a small brain area called the mirror system (the same area that lets a person feel a small stab of sadness when seeing another's knee hit a door), which means a middle-school student who catches a yawn easily is statistically more likely to be a student who gently notices other people's feelings — so the next time a

student catches five yawns in one afternoon she should not be embarrassed, she is simply a person whose mirror system is healthy and quietly awake.
B. Children under four catch yawns most easily and adults never catch them.
C. Catching a yawn is a sign of illness and poor social skills.
D. The mirror system has nothing to do with yawning or empathy.

4. What three ways of catching a yawn are described in Fact 4?

A. Yawning can only be caught by seeing somebody yawn in real life, never from a photo or sound.
B. Careful experiments have shown yawns can be caught without actually seeing a yawning person: a silent photograph of somebody yawning causes about 40% of viewers to yawn within one minute; a sound recording of a yawn with no video at all causes about 30% to yawn; and in one small 2003 study in Britain, merely reading the word "yawn" several times on a page caused 22% of readers to yawn, even alone in a silent room — the yawn reflex seems not tied to sight alone but to the idea of yawning itself, wherever that idea enters the brain.
C. Only children can catch yawns from silent photographs.
D. Yawns can be caught only by reading the word "yawn" aloud in public.

5. What does Fact 5 say about dogs and yawning?

A. Dogs can never catch yawns from humans because dogs are from a different species.
B. Dogs always catch yawns from strangers with the same rate as from their owners.
C. A 2013 study at the University of Tokyo showed that about 45% of domestic dogs yawn within a minute of watching their owner yawn, and the rate rises to roughly 60% when the owner is the dog's main carer; dogs do not catch yawns from strangers as easily, which is one small reason many scientists now believe that catching a yawn is at its heart a cross-species social signal — a quiet way living animals say, "I am paying attention to you, and my body is, in a small way, syncing with yours."
D. Domestic dogs catch yawns only from other dogs and never from humans.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段
【解析】"1987 Maryland 纯氧/高 CO_2 均不变 / 非关氧气 / 脑冷却"直引。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题

【定位】第③段
【解析】"Gallup 0.3–0.4℃ / 颚扩血管 / 湿壁 + 冷气 / 颈压热血回 / 小冷不深醒"直引。A、C、D 均错。故选 B。

3.
【答案】A
【题型】细节理解题
【定位】第④段
【解析】"8 国 60–70% / 4 岁以下少 / 镜像神经 / 共情体征 / 非弱点"直引。B、C、D 均错。故选 A。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段
【解析】"静照 40% / 声 30% / 英 2003 读词 22% / 关乎念头非视觉"直引。A、C、D 均错。故选 B。

5.
【答案】C
【题型】细节理解题
【定位】第⑥段
【解析】"东大 2013 / 45% / 主人 60% / 陌生人少 / 跨物种社交信号"直引。A、B、D 均错。故选 C。